

Vasileios Evangelopoulos

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 GitHub |  Website

EDUCATION

University of Patras

MEng in Electrical and Computer Engineering

October 2020 – October 2025

Patras, Greece

- GPA: 8.22/10.0
- 5 year program with integrated Master's degree
- **Thesis:** *Control design of HVDC link with virtual synchronous generator and implementation on real-time simulator*

PROJECTS

Control design of HVDC link with virtual synchronous generator and implementation on real-time simulator | *Matlab, Typhoon HIL, Python, PHIL*

[More info](#)

- Designed a VSM-based control architecture with inner current and outer voltage control loops
- Proposed an HVDC transmission system integrating two virtual inertia technologies
- Validated the system via a PHIL experimental setup using two Typhoon HIL real-time simulators, programmable bidirectional PSUs, and an HVDC line module

B-H Curve using Arduino | *Python, Matlab, Arduino, FEMM, Circuit Design*

[More info](#)

- Captured the hysteresis loop of a transformer core material using the Arduino UNO microcontroller
- Designed circuits for measuring primary winding current and secondary voltage
- XY plot using Matlab and Manim python library for visualization

FEA of a Reluctance Machine | *FEMM 4.2, Matlab, AutoCAD, Blender*

[More info](#)

- Designed a synchronous reluctance machine in AutoCAD
- Implemented a Matlab script to control FEMM software
- Post-processed FEMM data to calculate various performance parameters
- Designed a 3D model for visuals

Differential Drive Robot | *ROS, Python, Arduino, Linux*

[More info](#)

- Implemented state estimation for a differential drive robot using Dead Reckoning method
- Developed a localization system by implementing Adaptive Monte Carlo (AMCL) via Particle Filters
- Visualized real-time sensors data and system states using RViz
- Programmed firmware using Arduino

Monitoring & Control Propane Levels in a Tunnel | *LabVIEW, Arduino*

[More info](#)

- Implemented monitoring and control system using LabVIEW
- Developed a gas actuation mechanism and a tunnel apparatus

PUBLICATIONS

V. Evangelopoulos, M.-E. Karaitsi, T. Kavvathas, T. Alexandridis, G. C. Konstantopoulos

2025

- "Combined Synchronverter-VSM based HVDC transmission design with Power-Hardware-In-the-Loop (PHIL) implementation", 2025 IEEE PES Conference on Innovative Smart Grid Technologies - Middle East (ISGT Middle East), Dubai, United Arab Emirates, doi: 10.1109/ISGTMiddleEast65737.2025.11314231.

N. Bali, V. Evangelopoulos, M.-E. Karaitsi, P. Bouras and J. Gialelis

2025

- "Human-centric approach for determination of Thermal Plume theoretically and experimentally", 2025 IEEE 30th International Conference on Emerging Technologies and Factory Automation (ETFA), Porto, Portugal, doi: 10.1109/ISGTMiddleEast65737.2025.11314231.

TECHNICAL SKILLS

Programming Languages: Matlab/Simulink, Python, LabVIEW, C++, HTML/CSS

Software: FEMM, Typhoon HIL, AutoCAD, Multisim

Tools: Git, GitHub, Linux, L^AT_EX, ROS

VOLUNTEERING

Project Consultant

Mar 2023 - Jun 2023

Supervised and mentored a junior student team in the design and tuning of a PID controller for a ball-and-beam control system as part of an Electrical Engineering course project.

HONORS & AWARDS

Top 5 Power Engineering Diploma Dissertations in Greece Dec 2025

Dec 2025

Issued by 25th Workshop on Power Engineering Dissertations, IEEE Greek Power and Energy Society Chapter.